

Question				Marking details	Marks available					
					AO1	AO2	AO3	TOT	MATH	PRAC
9	(a)	(i)		Alveolus correctly labelled	1			1		
		(ii)		Any two (×1) from • large surface area (1) • thin <u>walls</u>/ one cell thick <u>wall</u> (1) • moist (lining) (1) • {rich/ close to/ good} blood supply (1)	2			2		
	(b)	(i)		{Cilia unable to move / less ciliated cells/ less cilia} and {Mucus has to be coughed out/ because mucus is not being pushed out/ because mucus stays in lung}			1	1		
		(ii)		Diameter of <u>airways</u> reduced/ smaller <u>airways</u> / narrower <u>bronchiole</u> (1) Less {gas exchange/ air able to move in and out / exchange of oxygen and carbon dioxide} (1)			2	2		
	(c)			The study is not valid because (must be present to award any marks but not creditworthy alone) There has only been one study (on e-flavourings)/ study has not been repeated/ not reproduced (1) only a {small number of/ 51} flavours have been studied/ sample size is small/ only 39 out of 7700 flavours contain diacetyl (1)			2	2		
				Total for question 9	3	0	5	8	0	0